MIL-C-17/22C
18 July 1985
SUPERSEDING
MIL-C-17/22B
9 October 1979

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE COAXIAL, PULSE, 48 OHMS, M17/22-RG27 (ARMORED), M17/22-00001 (UNARMORED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.

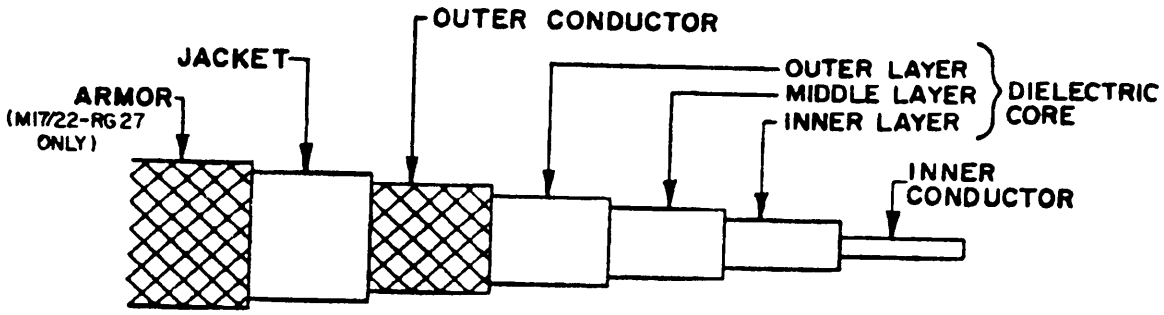


FIGURE 1. Configuration.

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TABLE I. Description.

Component	Construction details
Inner conductor	Nineteen strands of tinned, copper wire, each strand 0.0185 inch diameter. Overall diameter: .0925 inch ±.0030.
Dielectric core: Inner layer Middle layer Outer layer	Type D: Composite of three layers. Type H: Semiconducting synthetic rubber. Diameter: .133 inch ±.010. Type W: Insulating rubber. Diameter: .415 inch ±.020. Type H: Semiconducting synthetic rubber. Colored black. Diameter: .455 inch ±.010.
Outer conductor	Single braid of AWG No. 34, tinned, copper wire. Diameter: .500 inch maximum. Coverage: 93.1% nominal Carriers: 24 Ends: 10 Picks/inch: 9.5 ±10%
Jacket	Type IV: Diameter: .595 inch ±.015.
Armor (M17/RG27 only)	Single braid of aluminum-alloy wire. Diameter: .670 inch, maximum.

ENGINEERING INFORMATION:

Continuous working voltage: 12,000 V rms, maximum.

Operating frequency: 1 MHz, maximum.

Velocity of propagation: 49.5 percent, nominal.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): .191 ohm per 100 feet.

Elongation: 15 percent, minimum.

(C) Tensile strength: Not applicable.

Engineering note: This cable useful in pulse medium-low temperature applications (see connector per MIL-C-3607).

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

(C) Visual and mechanical inspection:

Out-of-roundness: Not applicable.

Eccentricity: Not applicable.

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 8 pounds, maximum.

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Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(C) Cold bend: -40°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: .062 inch, maximum.

Inner conductor from jacket: .125 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

(C) Flammability: Not applicable.

Weight: 33 pounds per 100 feet, maximum.

Electrical:

(C) Continuity: Applicable.

(C) Spark test: 6,000 Vrms, +10%, -0%.

(C) Voltage withstanding: 18,000 V rms, +10%, -0%.

Insulation resistance: 500 megohms, minimum.

Corona extinction voltage: 15,000 V rms, minimum.

Characteristic impedance: 48 ohms ±4.

Attenuation: 0.7 dB per 100 feet, maximum, at 1 MHz.

Structural return loss: Not applicable.

Capacitance: 55 pF per foot, maximum.

(C) Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

Part number		Superseded part number or type designation
Armored	M17/22-RG27	RG27A/U
Unarmored	M17/22-00001	

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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